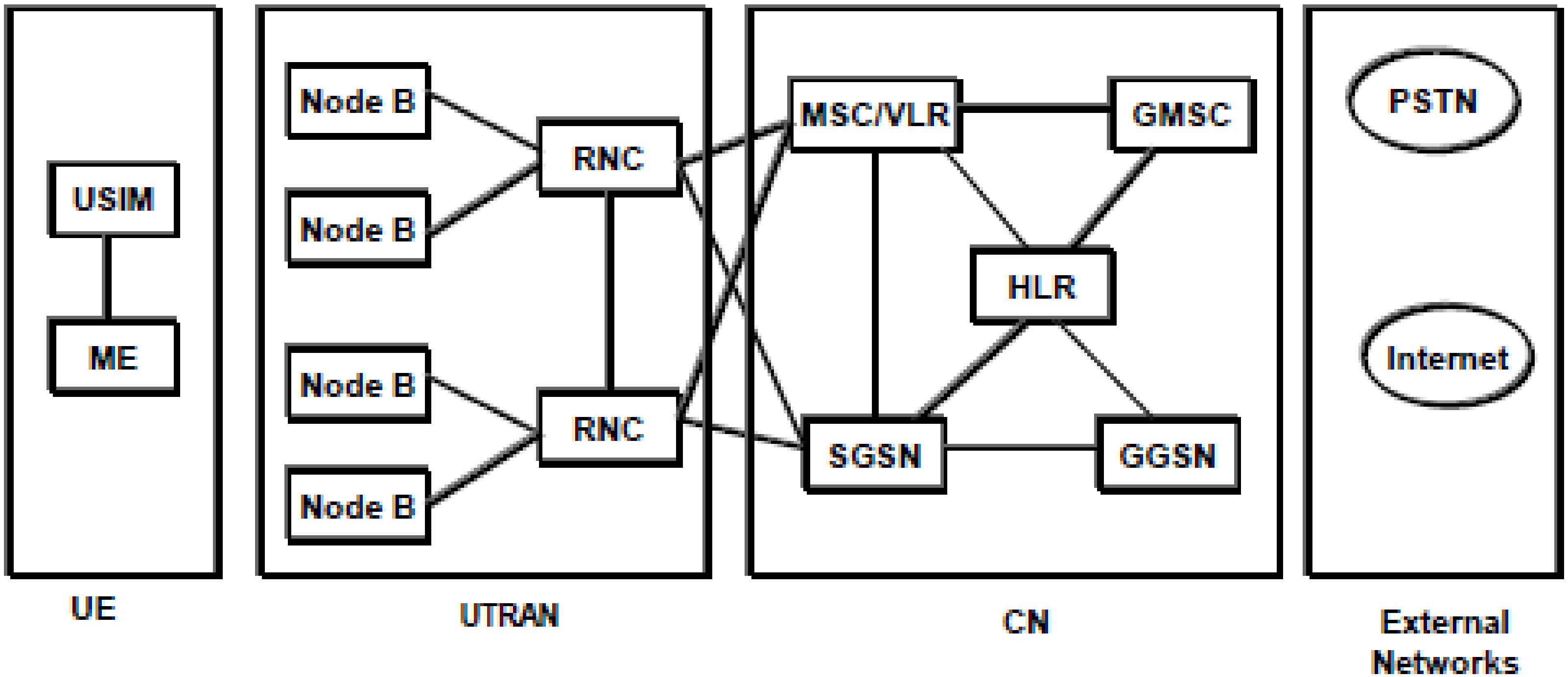
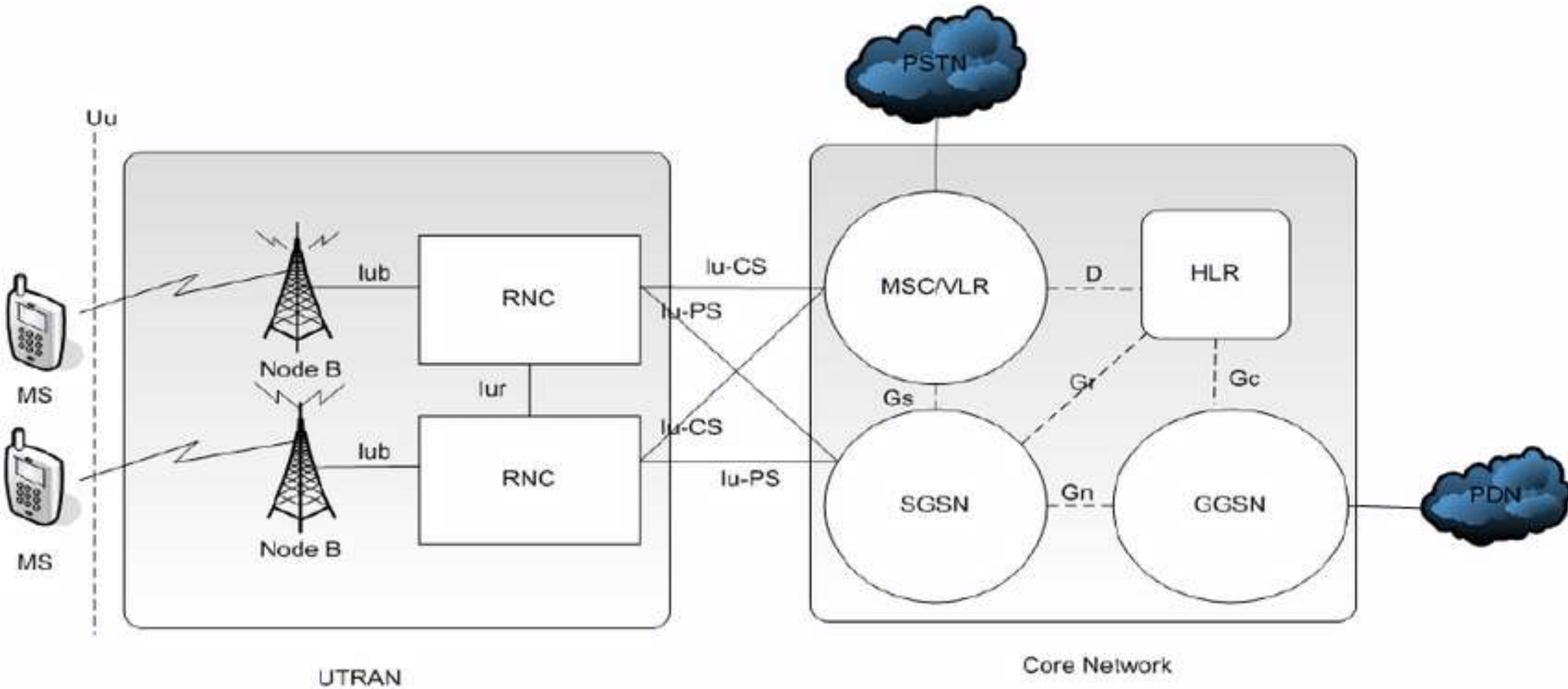
UMTS Architecture

UMTS System Architecture



- UE (User Equipment) that interfaces with the user
- UTRAN (UMTS Terrestrial Radio Access Network) handles all radio related functionality – WCDMA is radio interface standard here.
- CN (Core Network) is responsible for transport functions such as switching and routing calls and data, tracking users



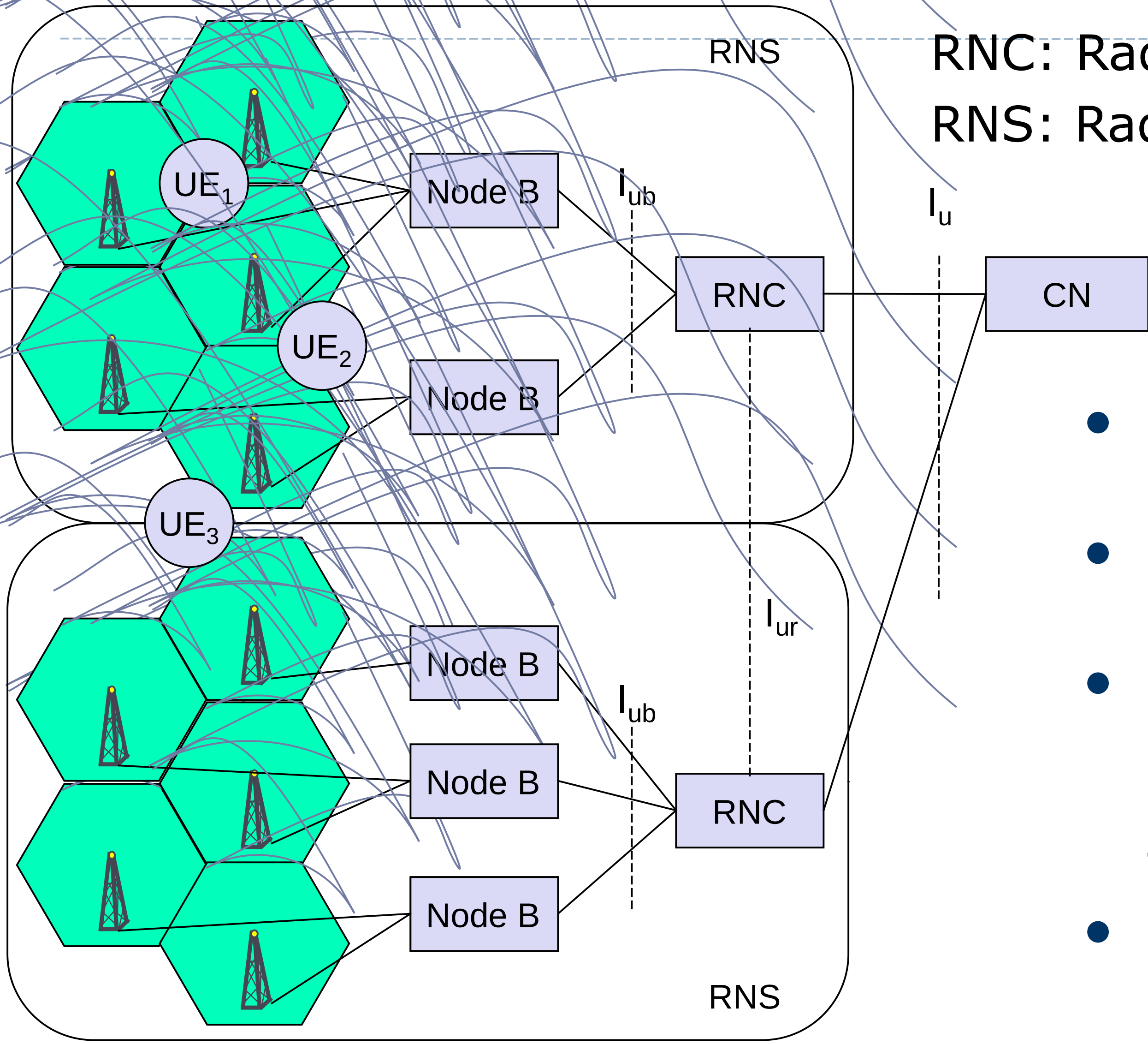
HLR: Home location register
MS: Mobile Station
PSTN: Public switched telephone network
SGSN: Serving GPRS support node
Node B: Base station
UTRAN: UMTS terrestrial radio access network

GGSN: Gateway GPRS support node
MSC: Mobile switching centre
RNC: Radio network controller
VLR: Visitor location register
PDN: Packet data network

UMTS System Architecture

- UE
  - ME (Mobile Equipment)
    - is the single or multimode terminal used for radio communication
  - USIM (UMTS Subscriber Identity Module)
    - is a smart card that holds the subscriber identity, subscribed services, authentication and encryption keys
- UTRAN
  - Node B (equivalent to BTS in GSM/GPRS)
    - performs the air interface processing (channel coding, rate adaptation, spreading, synchronization, power control).
    - Can operate a group of antennas/radios
  - RNC (Radio Network Controller) (equivalent to GSM BSC)
    - Responsible for radio resource management and control of the Node Bs.
    - Handoff decisions, congestion control, power control, encryption, admission control, protocol conversion, etc.

UTRAN architecture



RNC: Radio Network Controller

RNS: Radio Network Subsystem

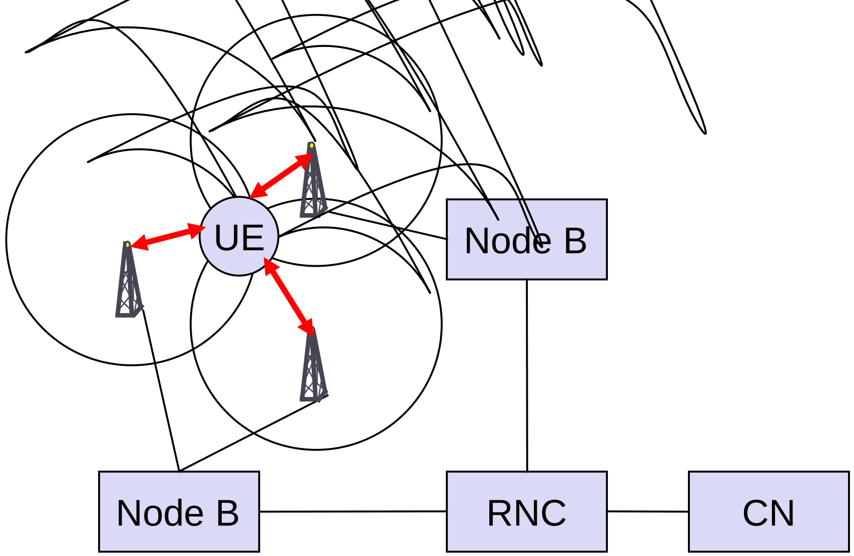
- UTRAN comprises several RNSs
- Node B can support FDD or TDD or both
- RNC is responsible for handover decisions requiring signaling to the UE
- Cell offers FDD or TDD

Core network

- The Core Network (CN) and thus the Interface I_u, too, are separated into two logical domains:
- Circuit Switched Domain (CSD)
  - Circuit switched service incl. signaling
  - Resource reservation at connection setup
  - GSM components (MSC, GMSC, VLR) are used
  - I_uCS interface
- Packet Switched Domain (PSD)
  - GPRS components (SGSN, GGSN)
  - I_uPS

- Release 99 uses the GSM/GPRS network and adds a new radio access!
  - Helps to save a lot of money …
  - Much faster deployment

Support of mobility: macro diversity



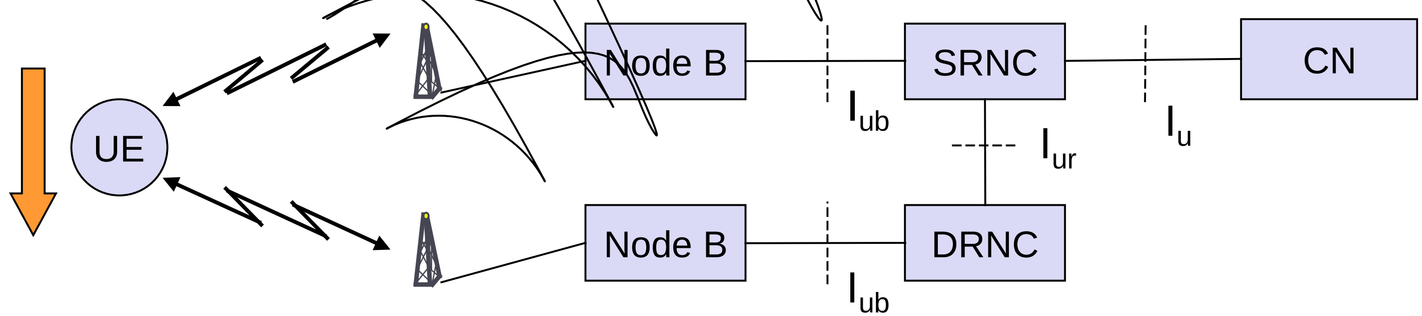
- Multicasting of data via several physical channels
  - Enables soft handover
  - FDD mode only
- Uplink
  - simultaneous reception of UE data at several Node Bs
- Downlink
  - Simultaneous transmission of data via different cells
  - Different spreading codes in different cells

UMTS – Rake Receiver

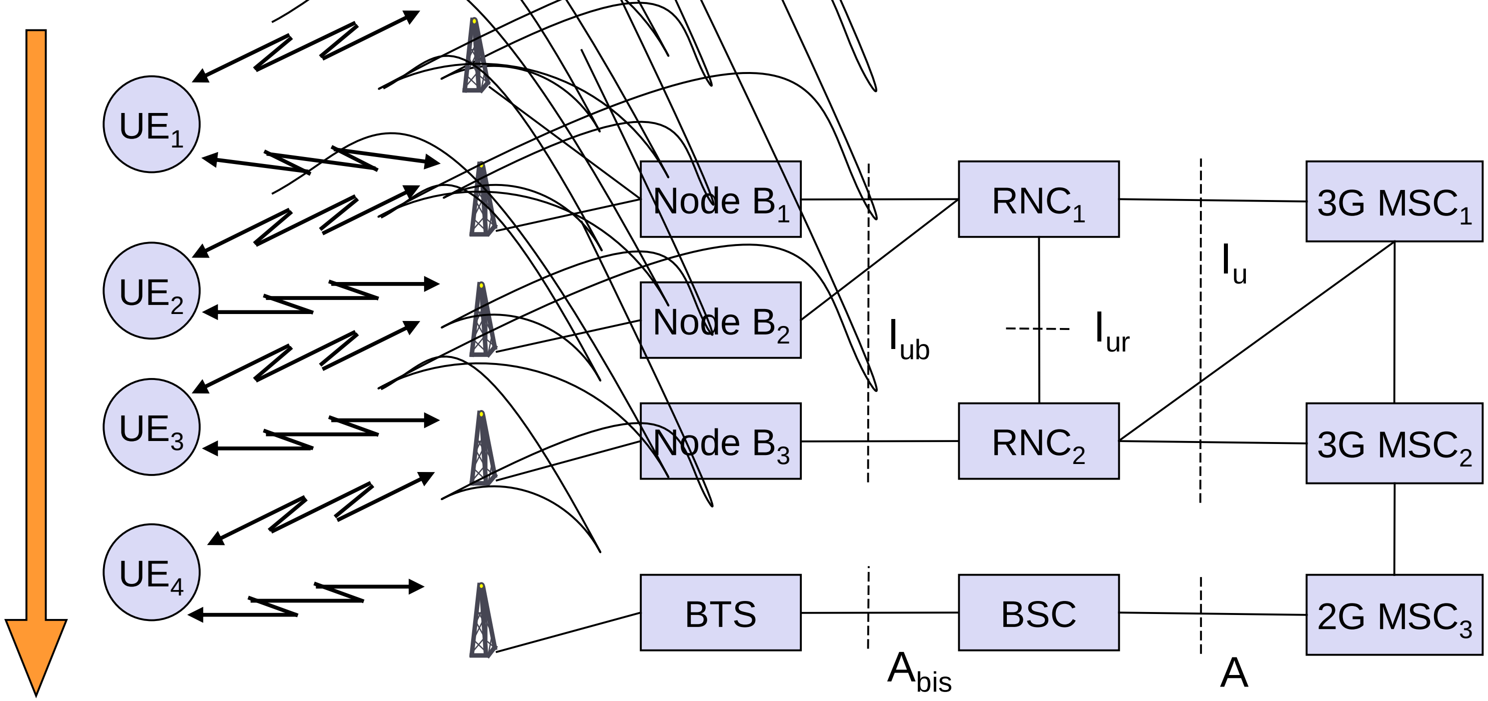
- Radio receiver
  - Uses several "sub-receivers" each delayed slightly in order to tune in to the individual multipath components
  - Each component decoded independently, but at a later stage combined in order to make the most use of the different transmission characteristics of each path
  - Results in higher Signal-to-noise ratio (or Eb/No) in a multipath environment than in a "clean" environment
  - Multipath fading is a common problem in wireless networks especially in metropoletan areas
- Another “trick” to increase connection quality and reliability is macro diversity

Support of mobility: handover

- From and to other systems (e.g., UMTS to GSM)
  - This is a must as UMTS coverage will be poor in the beginning
- RNS controlling the connection is called SRNS (Serving RNS)
- RNS offering additional resources (e.g., for soft handover) is called Drift RNS (DRNS)
- End-to-end connections between UE and CN only via I_u at the SRNS
  - Change of SRNS requires change of I_u
  - Initiated by the SRNS
  - Controlled by the RNC and CN



Example handover types in UMTS/GSM



Types of UMTS Handoffs

1. Intra RNC: between Node B's or sector of same Node B's attached to same RNC
2. Inter RNC: between Node B's attached to different RNC's, can be rerouted between RNC's locally if link , or rerouted by 3GMSC/SGSN, if RNC's in same service area
3. Inter 3GMSC/SGSN between Node B's attached to different
4. Inter System Handoff – between Node B and BTS along with a change of mode (WCDMA, GSM), (WCDMA, GPRS)

Note types 1,2, and 3 can be a Soft/Softer or Hard handoff, whereas, type 4 is always a Hard handoff

WCDMA

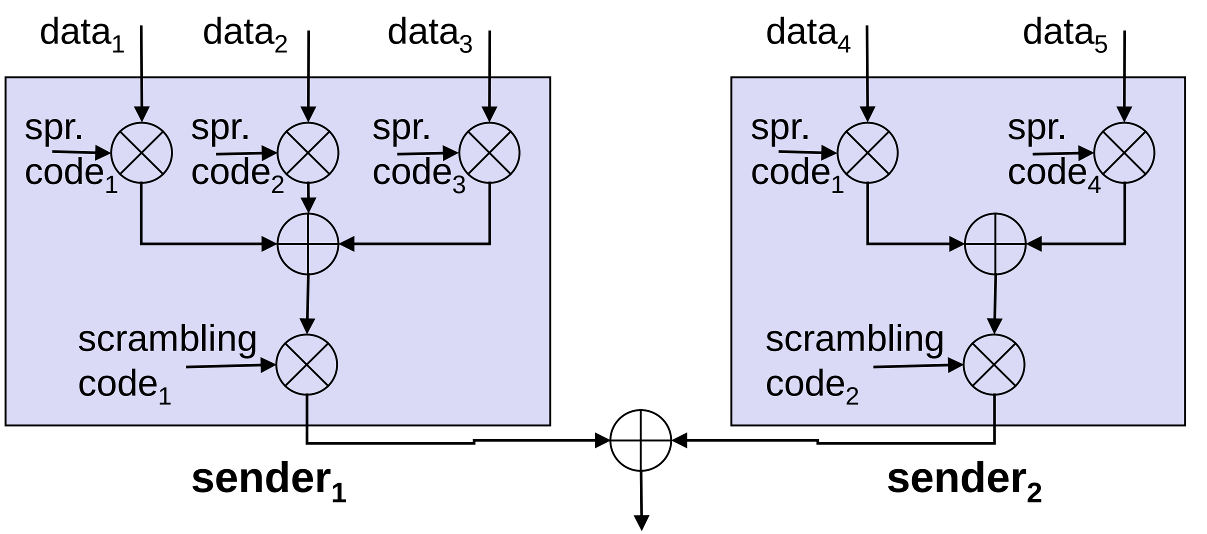
- Wideband Code Division Multiple Access (WCDMA)
  - The air radio interface standard for UMTS
  - Wideband direct sequence spread spectrum
  - Variable orthogonal spreading for multiple access (OVSF)
- Three types of interface：
  - FDD: separate uplink/downlink frequency bands with constant frequency offset between them
  - TDD: uplink/downlink in same band but time-shares transmissions in each direction
  - Dual mode :supports FDD and TDD
- Wide range of data rates due to CDMA with variable spreading, coding and modes
  - Varying user bit rate is mapped to variable *power* and *spreading*
  - Different services can be mixed on a single carrier for a user

Scrambling and Channelization

- Channelization codes are orthogonal codes
  - Separates transmissions from the same source
  - Uplink: used to separate different physical channels from the same UE – voice and data session
  - Downlink: used to separate transmissions to different physical channels and different UEs
  - UMTS uses orthogonal variable spreading codes
- Scrambling (pseudonoise scrambling)
  - Applied on top of channelization spreading
  - Separates transmissions from different sources
  - Uplink effect: separate mobiles from each other
  - Downlink effect: separate base stations from each other

Spreading and scrambling of user data

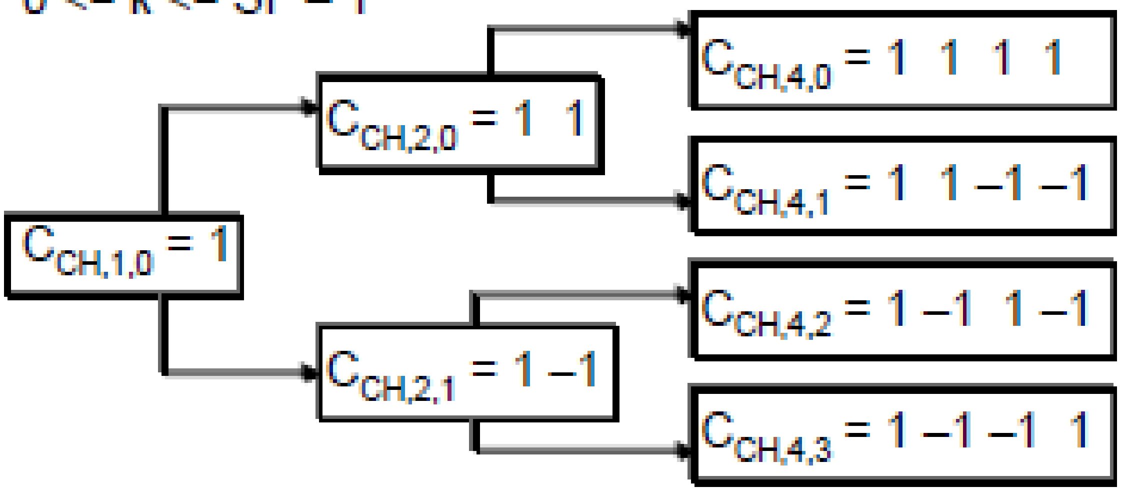
- Constant chipping rate of 3.84 Mchip/s
- Different user data rates supported via different spreading factors
  - higher data rate: less chips per bit and vice versa
- User separation via unique, quasi orthogonal scrambling codes
  - users are not separated via orthogonal spreading codes
  - much simpler management of codes: each station can use the same orthogonal spreading codes
  - precise synchronization not necessary as the scrambling codes stay quasi-orthogonal



WCDMA Variable Spreading

The channelization codes are Orthogonal Variable Spreading Factor codes that preserves the orthogonality between a user's different physical channels. The OVSF codes can be defined using a code tree.

In the code tree the channelization codes are uniquely described as $C_{CH,SF,k}$ where SF is the Spreading Factor of the code and k is the code number, $0 <= k <= SF – 1$



SF = 1 SF – 2 SF = 4 SF between 4 and 512 on DL

between 4 and 256 on UL

Breathing Cells

- GSM
  - Mobile device gets exclusive signal from the base station
  - Number of devices in a cell does not influence cell size
- UMTS
  - Cell size is closely correlated to the cell capacity
  - Signal-to-nose ratio determines cell capacity
  - Noise is generated by interference from
    - other cells
    - other users of the same cell
  - Interference increases noise level
  - Devices at the edge of a cell cannot further increase their output power (max. power limit) and thus drop out of the cell
    ⇨ no more communication possible
  - Limitation of the max. number of users within a cell required
  - **cell breathing** is a mechanism which allows overloaded cells to offload subscriber traffic to neighboring cells by changing the geographic size of their service area. Heavily loaded cells decrease in size while neighboring cells increase their service area to compensate. Thus, some traffic is handed off from the overloaded cell to neighboring cells, resulting in load balancing.